

CO₂ Partial Pressure-Steered Electrolysis for Seawater Resource Valorization and Carbon Sequestration

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Abstract: Atmospheric decarbonization is an essential engineering imperative for climate remediation. Although the ocean is a vast natural carbon sink capable of CO₂ mineralization via abundant divalent cations, existing electrochemical technologies still rely on broad pH modulation, lacking both the selective valorization of marine components and rigorous system-level validation, and thus remain confined to proof-of-concept stages. Here, we develop the photovoltaic-driven seawater electrolysis (PV-SE) system that integrates atmospheric CO₂ capture with the simultaneous co-production of green hydrogen, chlorine-based disinfectants, CaCO₃, and Mg(OH)₂ nanomaterials. A mechanistic study reveals a tunable, four-pathway mineralization process governed by the CO₂ partial pressure ($p\text{CO}_2$). This $p\text{CO}_2$ steering mechanism, transitioning the mineralization mechanism from traditional pH-mediated carbonate speciation to a $p\text{CO}_2$ -controlled interfacial steering pathway, enables the selective, phase-pure recovery of food-grade Mg(OH)₂ (99.43%) and CaCO₃ (99.78%). A 250-hour coastal integrated system demonstration achieved full energy self-sufficiency and a record sequestration capacity of 2,465.74 g CO₂ m⁻³. Techno-economic analysis projects a gross profit of \$104.12 per ton of CO₂ captured, suggesting a viable framework for coastal carbon management and the sustainable valorization of marine resources.

Introduction

Sustained anthropogenic carbon dioxide (CO_2) emissions remain the primary driver of global climate change^[1-4], posing a formidable challenge to ecosystems and human societies worldwide^[5-7]. Coastal urban cities, which house a majority of the global population and account for up to 70% of global CO_2 emissions^[8,9], represent a dual nexus on both major emission sources and primary targets for localized climate impacts^[10-12]. To mitigate these threats, carbon capture technologies have emerged as an essential strategy, ranging from industrial point-source capture^[13-15] to direct air capture (DAC)^[16-19]. However, conventional DAC technologies face significant thermodynamic and economic bottlenecks; amine- or hydroxide-based scrubbing systems rely on intensive thermal cycling for sorbent regeneration^[20,21], while emerging electrochemical approaches often require costly redox-active reagents^[22,23]. Consequently, a significant technological gap persists in the development of distributed, self-sufficient capture systems capable of valorizing captured carbon into high-value resources.

In contrast to terrestrial constraints, the ocean offers an expansive, chemically dynamic reservoir for carbon management, with a storage capacity of approximately 38,400 gigatons supported by abundant dissolved metal ions^[24,25]. While harnessing this marine reservoir has attracted significant traction^[26], existing electrochemical seawater carbon capture technologies generally fall into two categories: one focuses on adsorption-desorption cycles to enrich and recover CO_2 for subsequent carbon valorization^[27-30]; the other relies on mineralization with dissolved seawater cations (primarily Ca^{2+} and Mg^{2+}) to capture and sequester CO_2 ^[31-33]. Within the mineralization pathway, conventional strategies typically depend on the electrolytic modulation of seawater pH to specific thresholds to drive CO_2 speciation^[34-36], followed by the sequential recovery of calcium and magnesium precipitates based on their distinct solubility products (K_{sp})^[37,38]. Nevertheless, these methods necessitate continuous, high-precision pH feedback control to maintain stable mineralizing environments, introducing considerable operational complexity and system instability. Moreover, the interfacial nucleation-mineralization pathways and the intricate kinetic competition between Ca^{2+} and Mg^{2+} during real-time carbon capture remain poorly understood. Critically, the dynamic interplay between pH-mediated carbonate speciation and CO_2 supply equilibrium in governing the phase purity of recovered precipitates remains to be systematically elucidated. Elucidating the interfacial mechanisms governing mineralization and precipitation is essential for achieving synergistic, phase-pure recovery of marine resources and CO_2 capture.

Here, we demonstrate a photovoltaic-driven seawater electrolysis (PV-SE) system that integrates atmospheric CO_2 capture with full-spectrum recovery of marine resources. This energy-autonomous platform leverages a tunable, four-pathway carbon mineralization process that exploits interfacial interactions between dissolved cations (Ca^{2+} , Mg^{2+}) and anions (CO_3^{2-} , HCO_3^- , OH^-) in seawater. By strategically regulating the CO_2 partial pressure ($p\text{CO}_2$) within this system, we achieved the selective co-production of high-purity, food-grade calcium carbonate (CaCO_3 , 99.78%) and magnesium hydroxide ($\text{Mg}(\text{OH})_2$, 99.43%) nanomaterials, alongside the stable generation of high-purity green hydrogen (H_2) and chlorine-based disinfectant. We substantiate the engineering viability of this approach through a continuous 250-hour integrated system demonstration in natural seawater, which achieved full energy self-sufficiency and a record CO_2 fixation capacity of $2464.74 \text{ g CO}_2 \text{ m}^{-3}$. Furthermore, using in situ electrochemical X-ray diffraction (EC-XRD), we elucidate the atomic-level mechanisms governing alkalinity enhancement and the synergistic kinetics of mineral precipitation. Collectively, by transitioning the mineralization mechanism from traditional pH-mediated carbonate speciation to a $p\text{CO}_2$ -controlled interfacial steering pathway, this work establishes a self-sufficient, dual-engine framework that synergizes marine resource valorization with carbon mitigation.

Results and Discussion

Concept within this work

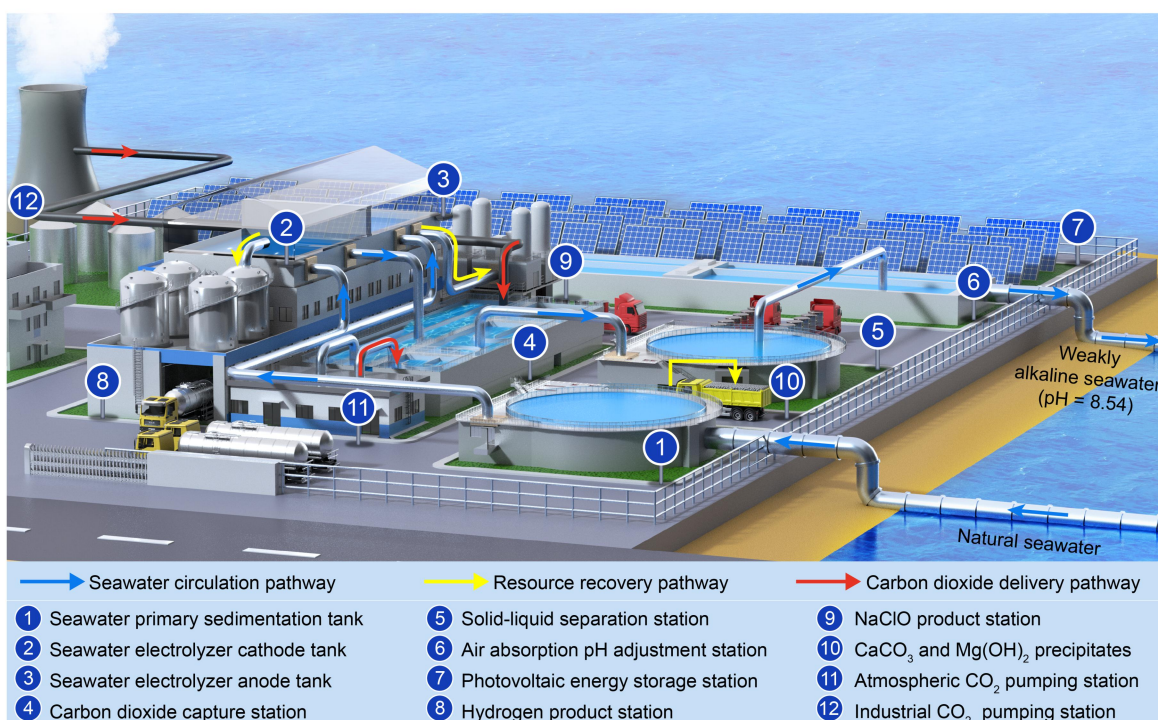
Modern coastal urban areas face a formidable dual challenge of mitigating the high CO_2 emissions stemming from dense industrial activity while simultaneously meeting the soaring global demand for critical mineral resources essential for infrastructure materials^[39], clean energy^[40], and public health^[41]. To strategically address this resource-emission nexus, we introduce an integrated PV-SE system (Figure 1a) designed to valorize the ocean's vast chemical inventory. This system synergistically produces high-purity Ca and Mg nanomaterials, green H_2 fuel, and chlorine-based disinfectants, all powered by solar energy while actively sequestering atmospheric CO_2 .

The system architecture centers on an integrated electrochemical seawater electrolyzer (Figure 1b). Raw seawater is filtered and sedimented to remove suspended solids and colloidal matter before entering the electrochemical

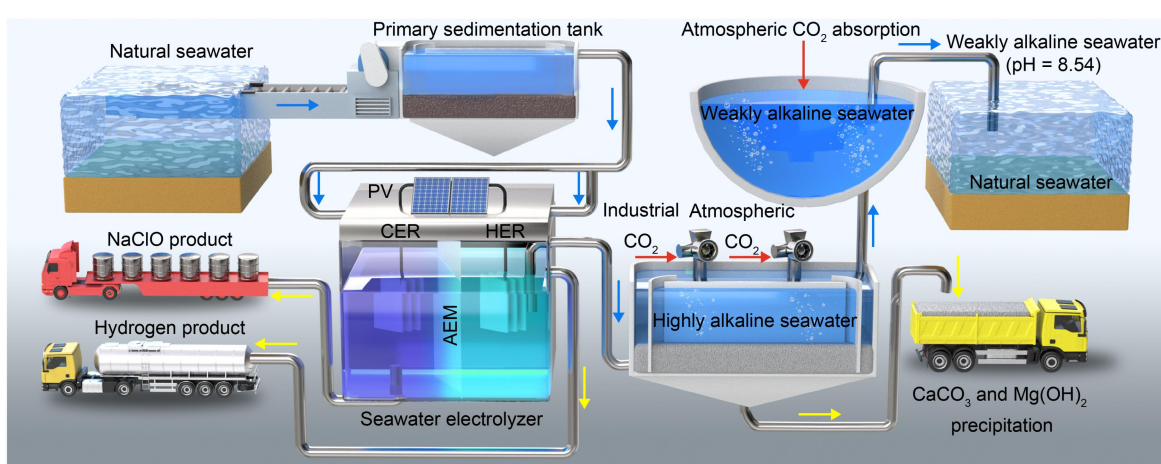
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cell. The electrolyzer configuration features a cathode for the hydrogen evolution reaction (HER) and a chlorine evolution reaction (CER) anode, separated by an anion exchange membrane (AEM). This design selectively promotes H_2 gas generation at the cathode, which substantially elevates the effluent seawater pH and alkalinity. Simultaneously, chloride oxidation at the anode produces sodium hypochlorite (NaClO) disinfectant solution. To optimize performance, the CER chamber operates under closed-loop circulation, while the HER chamber is continuously fed with fresh seawater, maintaining a controlled residence time to maximize both H_2 yield and alkalinity enhancement. Following electrolysis, the highly alkaline HER effluent is directed to a dedicated CO_2 capture station. The introduction of ambient air or industrial flue gas into this vessel initiates rapid carbon capture, driving the precipitation of Ca/Mg nanomaterials and enhancing the seawater's carbon storage capacity. We define this process as a four-pathway carbon mineralization framework (Figure 1c), encompassing dual solid-phase and dual liquid-phase sequestration routes: 1) direct solid-phase sequestration via CaCO_3 precipitation; 2) indirect solid-phase sequestration via $\text{Mg}(\text{OH})_2$ precipitation (potential for carbonation to MgCO_3); 3) Direct liquid-phase capture through total carbon (TC) accumulation in the electrolyte; and 4) indirect liquid-phase capture via pH-driven alkalinity enhancement. Specifically, the rapid OH^- generation at the cathode elevates the effluent pH (pathway 4), kinetically facilitates the sequential precipitation of CaCO_3 (pathway 1), which has a lower pH threshold for nucleation, followed by $\text{Mg}(\text{OH})_2$ (pathway 2) [35,42]. Concurrently, the increase in TC within the electrolyte reflects direct CO_2 absorption and carbonate formation mediated by OH^- (pathway 3). Furthermore, we demonstrate that pre-emptive stripping of dissolved CO_2 from seawater can effectively tune the precipitation order and purity of minerals. Upon solid-state separation, the resulting supernatant exhibits reduced alkalinity and a pH approaching that of ambient seawater as it continues to absorb CO_2 . This conditioned solution can be safely discharged, thereby enhancing in situ alkalinity and mitigating localized ocean acidification. Powered by a dedicated photovoltaic-storage unit, the PV-SE system operates as a fully closed-loop energy supply, providing a sustainable and economically viable solution for the valorization of marine resources.

a



b



c

Four-pathway carbon mineralization

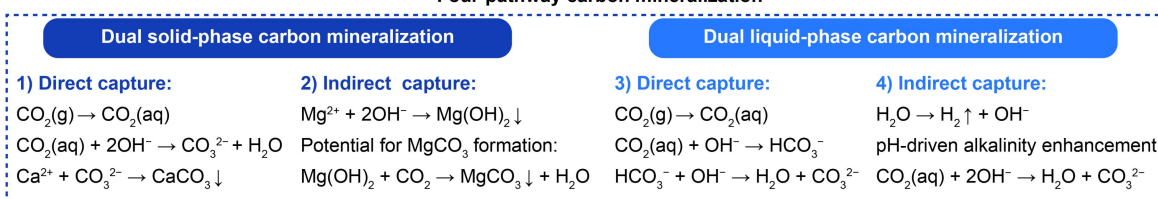


Figure 1. Conceptual architecture and mechanistic framework of the PV-SE system for coastal carbon neutrality. (a) Schematic illustration of a decentralized coastal industrial park integrated with the photovoltaic-driven seawater electrolysis (PV-SE) system for localized CO_2 mitigation. (b) Comprehensive material flow and mass balance within the integrated electrolyzer system. (c) Four-pathway carbon mineralization framework for the carbon capture process in the PV-SE system.

Seawater Electrolysis Performance and Divergent Resource Recovery Kinetics

To evaluate the electrochemical efficacy of the proposed system, we first characterized its performance using an H-type electrolyzer configuration (Figure 2a). Commercial Pt/C and RuO_2 -coated titanium electrodes were selected as cathode and anode, respectively, owing to their known robustness toward HER and CER activity in highly saline media^[43,44]. Linear scan voltammetry (LSV) in a standard three-electrode setup (Figure S1) confirmed their superior electrochemical profiles (Figure 2b). Specifically, the Pt/C cathode achieved a favorable HER potential of -0.25 V

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(vs. RHE) at a practical current density of 10 mA cm^{-2} . Notably, the overpotential exhibited an inverse correlation with increasing alkalinity, suggesting a self-optimizing performance profile in the alkalinity-enhanced HER chamber effluent. Concurrently, the RuO_2 anode exhibited competitive selectivity in natural seawater, where the CER was initiated at 1.89 V (vs. RHE) at 10 mA cm^{-2} . This effectively outcompeted the energetically less favorable oxygen evolution reaction (OER, 2.08 V vs. RHE at 10 mA cm^{-2} in $0.50 \text{ M Na}_2\text{SO}_4$ reference electrolyte). Furthermore, the CER overpotential decreased with increasing local acidity. This unique pH-dependent energy consumption profile, characterized by reduced overpotential under conditions of increased local alkalinity at the cathode and acidity at the anode, provides compelling evidence that continuous operation inherently enhances the system's overall thermodynamic efficiency.

Building on these fundamental insights, the two-electrode system (Figure S2a), exhibited an initiation potential of 1.45 V , reaching practical current densities of 10 and 100 mA cm^{-2} at 2.69 V and 6.76 V , respectively (Figure S2b). Under constant current operation at 10 mA cm^{-2} , the dual-chamber configuration achieved stable, simultaneous co-production of chlorine-based disinfectants and green H_2 : In the anode chamber, free chlorine accumulated steadily, accompanied by H^+ generation via the OER side reaction, reaching a stable concentration of 115.9 mg L^{-1} at a terminal pH of 1.95 (Figures 2c and S2c), thereby facilitating on-site disinfectant generation for aquaculture use. In the cathode chamber, H_2 was generated at a stable rate of 58.11 mL h^{-1} via uninterrupted HER, concurrently driving a critical pH elevation to 10.15 (Figures 2d and S2d). This cathodic alkalinity was immediately harnessed for mineral recovery (Pathways 1 and 2) and carbon capture (Pathways 3 and 4). Upon continuous introduction of CO_2 -containing ambient air into the cathode chamber via a microporous gas diffuser during electrolysis, we observed a distinct kinetic dichotomy in the removal of Ca^{2+} and Mg^{2+} (Figures 2e and S2e-f). The uninterrupted supply of CO_2 and OH^- led to a sharp, near-linear attenuation of Ca^{2+} concentration, achieving an ultimate removal efficiency of 99.7% . In contrast, removal of Mg^{2+} exhibited a distinct induction period, with only 16.4% removed after the first 120 minutes due to the competitive precipitation with Ca^{2+} ; followed by a rapid linear decrease as the cumulative OH^- flux exceeded the competition threshold, eventually reaching 61.2% within the test period. This kinetic lag in magnesium removal provides *prima facie* evidence for a preferential precipitation sequence dictated by the specific thermodynamic pH thresholds for CaCO_3 ($\text{pH} > 8.3$) versus $\text{Mg}(\text{OH})_2$ ($\text{pH} > 9.1$) in natural seawater (Figure S3) [35,36]. These results validate the proposed dual solid-phase capture mechanism within our four-pathway framework. XRD analysis of the cathodic solids confirmed a high-purity mixture of CaCO_3 and $\text{Mg}(\text{OH})_2$ (Figure S4a), while a 4.15 -fold increase in TC content within the alkaline electrolyte (Figure S4b), substantiated the dual liquid-phase capture mechanism. Collectively, these findings demonstrate that the PV-SE system effectively integrates carbon fixation with high-value mineral recovery and clean fuel production.

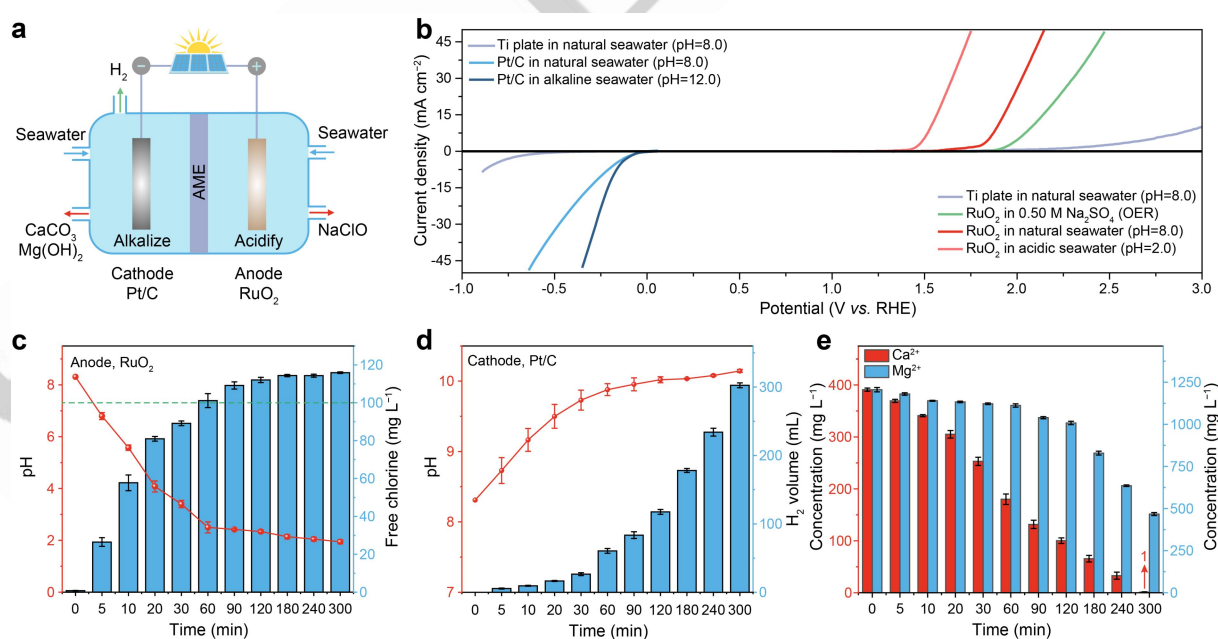


Figure 2. Evaluation of seawater electrolysis performance and resource recovery kinetics in an H-type two-electrode system. (a) H-type seawater electrolyzer configuration. (b) HER and CER polarization curves of Pt/C and RuO₂ in natural seawater. (c) Time-dependent pH and free chlorine evolution in the anode compartment, the green dashed line indicates the Chinese standard (GB28233-2020) lower threshold of available chlorine (predominantly as free chlorine) for disinfection. (d) Time-dependent pH and hydrogen evolution in the cathode compartment

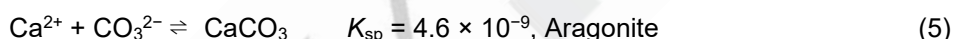
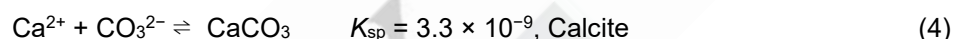
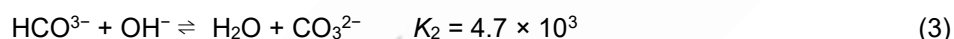
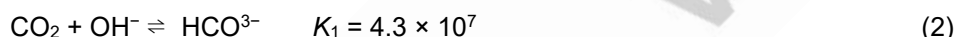
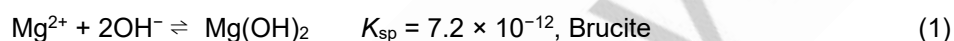
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(e) Time-dependent Ca^{2+} and Mg^{2+} concentration in the cathode compartment. 300 mL of natural seawater was used in each electrode compartment, the electrolytic current density is 10 mA cm^{-2} , and the error bars are means $\pm$ standard deviation of three replicates.

$p\text{CO}_2$ -Steered Dual Solid-Phase Mineralization Reaction Dynamics

To resolve the reaction dynamics of Ca and Mg dual solid-phase mineralization during seawater carbon capture, we further employed in situ EC-XRD, alongside ex situ scanning electron microscopy (SEM) and high-angle annular dark-field scanning transmission electron microscopy (HAADF-STEM) to probe the precipitate structure and the evolution of Ca/Mg speciation. We strategically explored the thermodynamic and kinetic landscape across three distinct CO_2 availability regimes: (i) abundant CO_2 supply (continuous aeration), (ii) insufficient CO_2 supply (limited to natural dissolved CO_2), and (iii) CO_2 -deficient conditions (achieved via N_2 stripping).

Under abundant CO_2 supply conditions, in situ EC-XRD data collected with a customized electrochemical cell (Figures 3a and S5) confirmed the simultaneous and rapid coprecipitation of both $\text{Mg}(\text{OH})_2$ and CaCO_3 (Figure 3b). Mechanistically, the OH^- generated at the cathode is partitioned into two parallel competitive pathways: 1) Direct hydroxyl coupling: OH^- reacts with Mg^{2+} to form Brucite-phase $\text{Mg}(\text{OH})_2$ via equation (1); 2) Carbonate mediated mineralization: OH^- facilitates CO_2 hydration to form CO_3^{2-} (equations (2-3)), which subsequently precipitates with Ca^{2+} to precipitate Calcite- or Aragonite-phase CaCO_3 (equations (4-5))^[37,38].



SEM and HAADF-STEM imaging with EDS mapping (Figures S6 and S11) revealed a complex composite architecture consisting of layered $\text{Mg}(\text{OH})_2$ nanosheets intermixed with CaCO_3 nanocubes/nanorods (Ca/Mg atomic ratio $\approx 1.5:1$). Integrating these findings with the operando data from the H-type electrolyzer experiments (Figures 2e and S2e,f), we reconstructed the electrolyte speciation and CO_2 partial pressure dynamics during the dual solid-phase mineralization process (Figure 3c), an end-up mixed solid with 40.03% $\text{Mg}(\text{OH})_2$ and 59.05% CaCO_3 (Figure 3d and Table S2), can be harvested.

Under insufficient CO_2 supply, mineralization shifts to a distinct sequential pathway governed by the temporal accumulation of cathodic OH^- , with the competitive precipitation regulated by bulk K_{sp} values and macroscopic pH thresholds. In situ EC-XRD analysis (Figure 3e) resolved this into two discrete stages: stage I (0-1.28 h) was dominated by the precipitation of CaCO_3 , during which limited dissolved CO_2 was rapidly consumed by the newly generated OH^- and Ca^{2+} . HAADF-STEM analysis corroborated this hypothesis, showing CaCO_3 nanocubes/nanorods with a high Ca/Mg ratio of 5.5:1 (Figure S7a). As electrolysis proceeded, stage II (1.28-3h) commenced as OH^- concentration accumulated further, surpassing the higher pH threshold required for $\text{Mg}(\text{OH})_2$ precipitation. Concurrently, the Ca/Mg ratio inverted sharply to 1:50 (Figure S7b), indicating $\text{Mg}(\text{OH})_2$ nanosheets as the overwhelmingly dominant phase. H-type electrolyzer experiments double corroborated these dynamic shifts: Ca^{2+} removal plateaued after 30 minutes in stage I (max 28% removal), while Mg^{2+} removal remained negligible in stage I (6%) before sharply increasing to 66% removal efficiency in stage II (Figure S8a-c). Electrolyte CO_2 partial pressure profiles visually indicate that Ca^{2+} reacts with dissolved CO_2 before Mg^{2+} participates (Figure 3f). Characterization of the precipitates confirmed a distinct compositional transition, shifting from CaCO_3 -dominant (86.57%) in stage I to $\text{Mg}(\text{OH})_2$ -dominant (82.92%) in stage II (Figure 3g and Tables S3, S4). Finally, subsequent atmospheric exposure of the residual alkaline seawater (simulating stage III) achieved complete Ca^{2+} removal via $\text{Mg}(\text{OH})_2$ -buffered CO_2 capture. XRD confirmed the phase evolution across all three stages (Figure S8d).

Under the CO_2 -deficient conditions via N_2 stripping, the system exhibits high selectivity for high-purity food-grade $\text{Mg}(\text{OH})_2$. In situ EC-XRD analysis (Figure 3h) and HAADF-STEM analysis (Figure S9a) verified that the precipitate is almost exclusively Brucite-phase $\text{Mg}(\text{OH})_2$, with negligible CaCO_3 formation. Quantitative analysis of the precipitate recovered in stage I (under N_2) confirmed selectivity, achieving 87% Mg^{2+} removal efficiency compared to only $1.5 \pm 0.78\%$ for Ca^{2+} (Figure S10a-c), a trend further corroborated by the partial pressure composition analysis of the electrolyte (Figure 3i). Quantitative analysis confirmed a high purity of 99.43% for $\text{Mg}(\text{OH})_2$, reaching food-grade quality (Figure 3j and Table S5). Critically, the subsequent injection of air into the alkaline supernatant

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(stage II) enabled complete Ca^{2+} removal (100% efficiency) and the crystallization of calcite-phase CaCO_3 nanocubes with a food-grade purity of 99.78% (Figure S9b and Table S6), demonstrating effective CO_2 capture and phase-pure resource recovery. XRD confirmed the presence of pure $\text{Mg}(\text{OH})_2$ in stage I and pure CaCO_3 in stage II (Figure S10d). This multi-stage purity-phase recovery, which transcends the constraints of conventional solution equilibration or static pH manipulation, is fundamentally driven by the interplay between electrochemical alkalinity propagation and the controlled supply of $p\text{CO}_2$. Within this framework, the $p\text{CO}_2$ supply conditions serve as a key regulatory factor governing the nucleation pathways of CaCO_3 and $\text{Mg}(\text{OH})_2$, respectively.

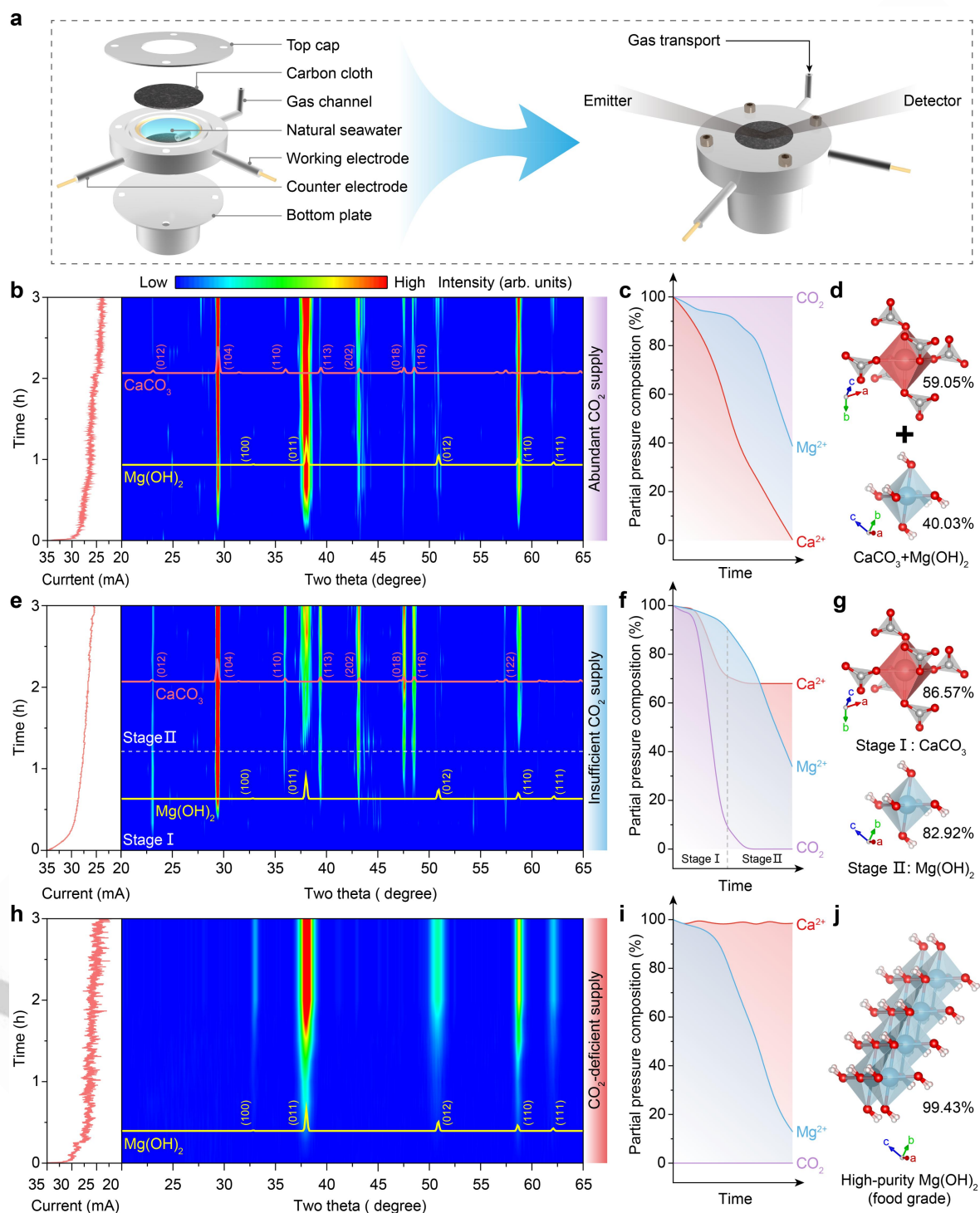


Figure 3. Reaction dynamics of $p\text{CO}_2$ -steered carbon mineralization in natural seawater. (a) Schematic representation of the custom-designed electrochemical cell for EC-XRD characterization. (b) Time-resolved in situ EC-XRD, (c) corresponding electrolyte speciation dynamics, and (d) precipitate crystal structures under abundant CO_2 supply conditions. (e) Time-resolved in situ EC-XRD, (f) electrolyte speciation dynamics, and

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(g) precipitate crystal structures under insufficient CO₂ conditions. (h) Time-resolved in situ EC-XRD, (i) electrolyte speciation dynamics, and (j) precipitate crystal structures under CO₂-deficient conditions.

CO₂ Partial Pressure-Steered Selective Mineralization Mechanism

To further elucidate the selective mineralization mechanism of the *p*CO₂-steered process, we employed theoretical simulations to clarify the precipitation boundaries of phase-pure minerals across pH gradients (Figure 4a-c and Note S1). The results demonstrate that high-purity CaCO₃ can be selectively precipitated within a narrow pH window of 8.3 to 9.1, provided a carbon source is present. In contrast, the recovery of phase-pure Mg(OH)₂ is contingent upon the complete depletion of dissolved carbon. Precipitation occurs only when the local pH surpasses the threshold for Mg(OH)₂ formation (pH ≥ 9.1 under CO₂-deficient conditions). Specifically, under abundant CO₂ conditions, precipitation requires a pH ≥ of 9.5, whereas under insufficient CO₂ conditions, it requires a pH ≥ of 9.7. These boundaries reflect the competitive equilibrium mechanism between hydroxyl-driven Mg²⁺ precipitation and carbonate-mediated Ca²⁺ sequestration. In short, our findings establish a significant departure from conventional methodologies that rely on complex, external pH titration^[35]. Instead, we demonstrate that strategic *p*CO₂ control, coupled with a staged and selective recovery process over the reaction-time (Figure 3b-j) dimensions, can serve as a "tuning fork" for selective mineral crystallization (Figure 4d-f). This temporally or spatially phased crystallization framework, driven by *p*CO₂ regulation, bypasses the tedious post-treatment separation of high-purity CaCO₃ and Mg(OH)₂ with potential large-scale operations, providing a practical route for seawater resource valorization and CO₂ capture.

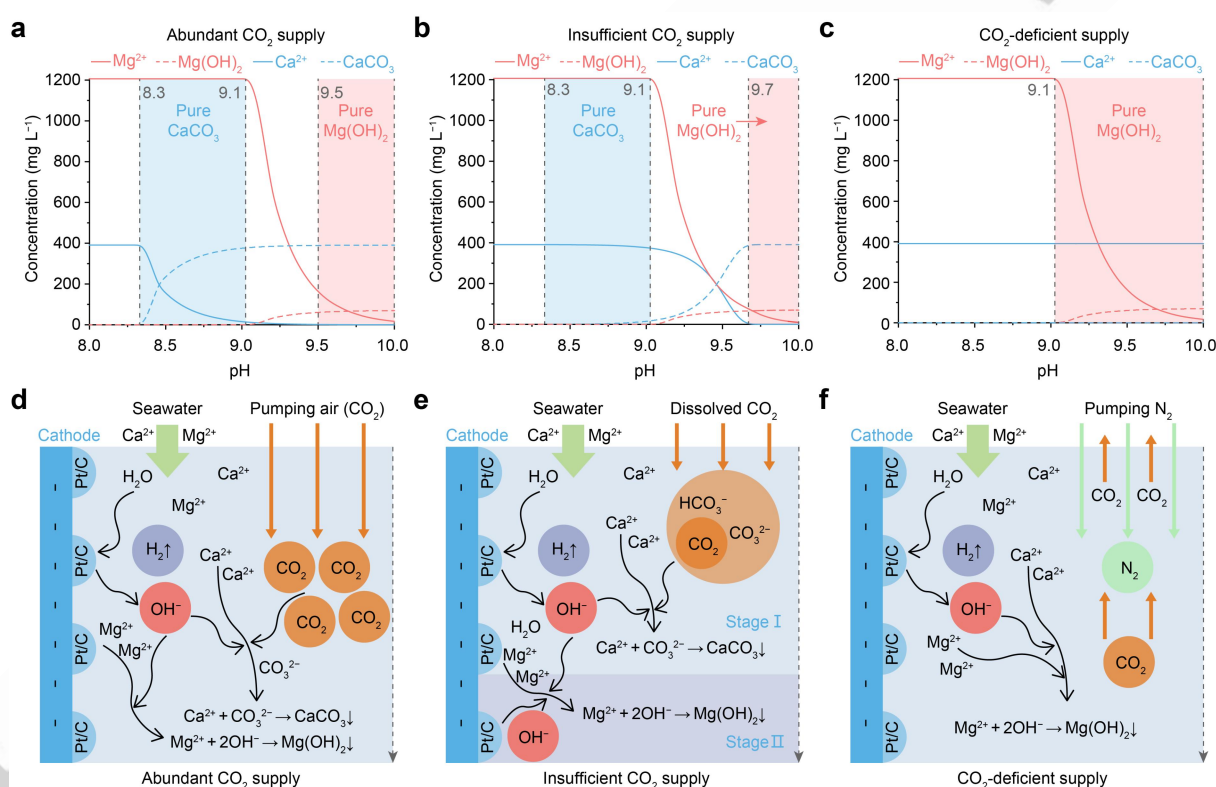


Figure 4. CO₂ partial pressure-steered selective mineralization mechanisms. Phase-pure precipitation boundaries in the pH dimension under (a) abundant CO₂ supply, (b) insufficient CO₂, and (c) CO₂-deficient conditions. Schematic illustration of the dual solid-phase mineralization and carbon capture mechanism within the electrolyzer under (d) abundant CO₂ supply, (e) insufficient CO₂, and (f) CO₂-deficient conditions.

Continuous-Flow Seawater Electrolysis System for Resource Recovery

To evaluate the practical viability of the seawater electrolysis system for industrial-scale deployment, we designed and fabricated a compact, continuous-flow seawater electrolyzer (dimensions: 5 cm × 5 cm × 1.1 cm) (Figures 5a and S12a). This compact device integrates seawater inlet/outlet, H₂ collection, CO₂ capture, and Ca/Mg precipitation recovery functions within the cathode chamber, enabling high spatial scalability. Compared to the H-type electrolyzer, the larger AEM area and smaller electrode distance in the continuous-flow electrolysis device reduce solution ohmic resistance and overpotential^[34], which translates to a substantially enhanced current

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response (Figure 5b) and stable current-voltage (J - V) behavior across both constant current and constant voltage operation modes (Figure S12b), thereby confirming its adaptability to fluctuating solar energy inputs.

In continuous-flow operation, the adequate reaction time, quantified by the reaction hydraulic retention time (RHRT) per unit volume, was identified as a critical parameter controlling mineralization kinetics. Under a current density of 10 mA cm^{-2} and an abundant CO_2 supply, an optimal RHRT of 20 minutes was required to achieve 100% Ca^{2+} removal (Figure 5c). Long-term stability tests under these optimized conditions demonstrated simultaneous and efficient sequestration of both Ca^{2+} and Mg^{2+} (Figures 5d and S12c-d). This kinetic pattern reinforces the preferential precipitation sequence observed in the batch-mode: Ca^{2+} removal reached 100% rapidly, while Mg^{2+} removal followed a linear trajectory, achieving a terminal efficiency of 80%. While high-purity $\text{Mg}(\text{OH})_2$ and CaCO_3 command greater commercial value, mixed precipitation offers greater CO_2 capture capacity because it continuously generates alkalinity. Meanwhile, high-purity H_2 was produced at a rate of 239.5 mL h^{-1} (Figure S12e-f), with a Faraday efficiency (FE) increasing from 64.11% to 96.11% (Figure S12g). This FE improvement is consistent with the enhanced HER kinetics at higher pH observed in the LSV tests, indicating that the system transitions into an optimally stable regime after initial operation. Simultaneously, the anode chamber produced disinfectant-grade free chlorine at a high concentration of $534 \pm 4.3 \text{ mg L}^{-1}$ (Figure S12h). To accurately resolve the transient product distribution during this process, chromatographic and speciation analyses were performed over a one-hour evaluation window collected within the anode chamber (Figures S13-S15), revealing FE of 32.5%, 62.42%, and 1.79% for the OER, CER, and bromine evolution reaction (BER), respectively. Throughout the 10-hour electrolysis process, the voltage remained stable within a narrow range of 2.62-2.83 V (Figure 5e), with a minor positive drift attributed to the inevitable accumulation of mineral precipitates on the cathode surface.

To ensure long-term operational stability and mitigate the performance decay caused by cathodic scaling, we implemented an advanced half-wave rectification strategy. This involved periodically reversing the current direction to dissolve and remove locally accumulated precipitates^[45]. By cycling between 10-hour electrolysis periods and 1-hour reverse-current recovery periods, the cell voltage was consistently restored. Over 10 complete cycles, the average working and recovery voltages were maintained at 2.64 V and -2.77 V, respectively (Figure 5f). Crucially, this robust operational strategy allowed the continuous-flow device to achieve stable operation for over 110 hours (Figure 5g), requiring a total cumulative energy consumption of 72.40 Wh (comprising 65.48 Wh for electrolysis and 6.92 Wh for descaling), which represents a 21.09% energy reduction compared to the 91.75 Wh consumed by the conventional continuous operation baseline projected from the average voltage growth rate of 0.021 V h^{-1} (Note S2). Furthermore, the efficacy of the periodic current reversal scheme in removing mineral scale was confirmed by physical characterization of the cathode surface (Figure S16). Specifically, SEM-EDS mapping detected only minimal residual Ca and Mg signals, while the active platinum catalyst layer remained structurally intact on the substrate without mechanical delamination. Throughout this extended demonstration, the system maintained a stable alkaline effluent ($\text{pH} = 9.95$), consistent 100% Ca^{2+} removal, and a reliable Mg^{2+} removal efficiency of $69.66 \pm 1.47\%$. These results definitively substantiate the engineering feasibility of the PV-SE concept for sustained real-world carbon capture and marine resource valorization.

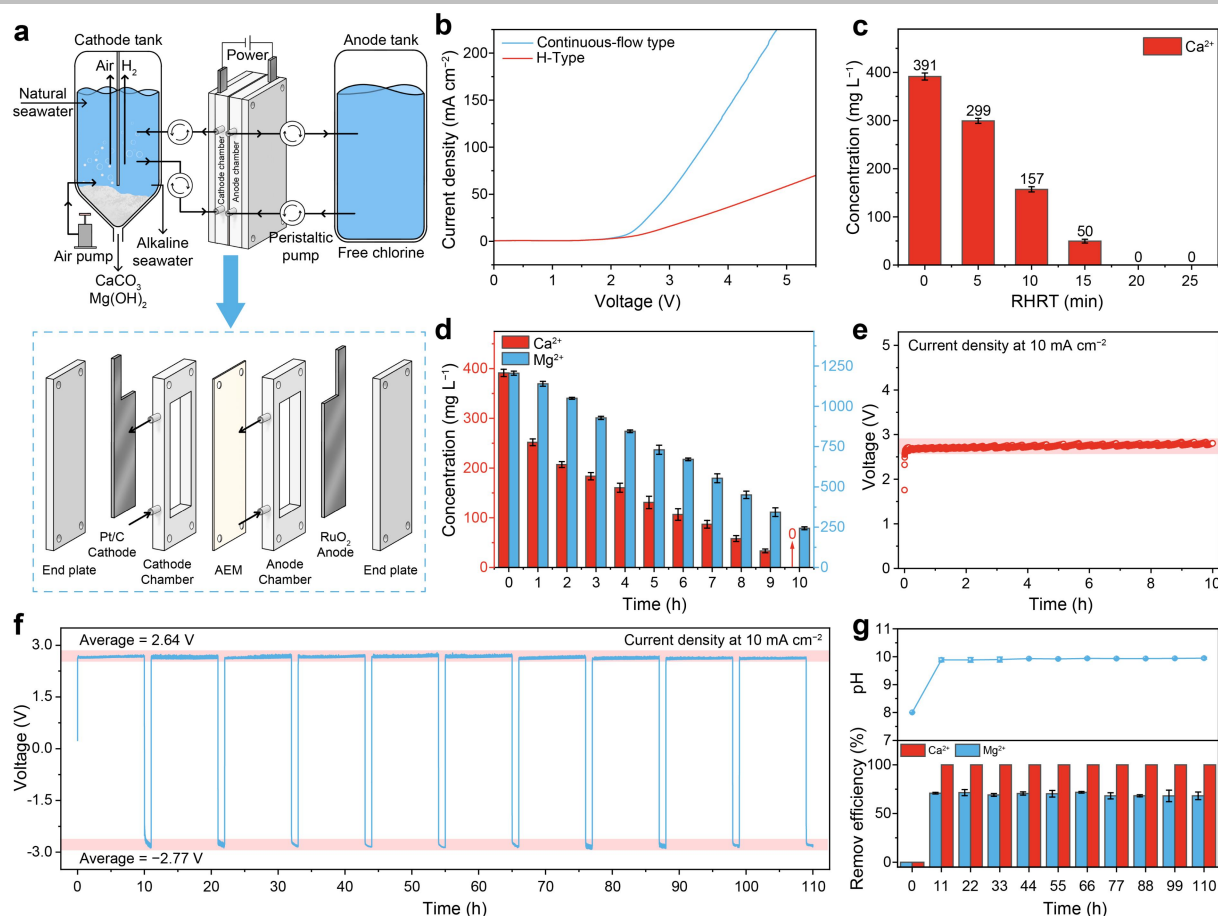


Figure 5. Performance and operational stability of the continuous-flow seawater electrolysis system. (a) Schematic architecture of the compact continuous-flow device. (b) Comparative J-V curves illustrating the superior current response of the continuous-flow system over the H-type electrolyzer. (c) Optimization of RHRT on Ca^{2+} removal. (d) Time-dependent Ca^{2+} and Mg^{2+} concentration profiles in the cathode effluent at 10 mA cm^{-2} and 20 min RHRT. (e) Voltage stability during a continuous 10-hour operational cycle. (f) Voltage variation and (g) Ca^{2+} and Mg^{2+} removal efficiency and pH fluctuations over 10 cycles of the half-wave rectification strategy. Error bars are means $\pm$ standard deviation of three replicates.

Coastal Validation and Performance of the Integrated PV-SE System

The synergistic integration of renewable solar energy with seawater-based carbon capture provides a transformative and sustainable pathway toward global carbon neutrality. To demonstrate the scalability, operational efficiency, and processing capacity of our system, we architected a fully integrated, multichannel PV-SE system for coastal validation (Figures 6a and S17a). The system architecture comprises a 400 W solar array, a 5.0 kWh battery storage unit, a four-channel continuous-flow seawater electrolyzer module, a constant-voltage power supply, and precision data acquisition units for monitoring H_2 evolution and mineral recovery. The coupling of PV generation with battery storage (Figure S17b) facilitates 24-hour continuous operation, a prerequisite for future industrial-scale deployment.

The integrated system was operated at a constant current density of 10 mA cm^{-2} with an RHRT of 20 minutes. To mitigate cathodic scaling during extended operation, we employed the previously validated half-wave rectification strategy (24 h of electrolysis followed by 1 h of reverse current). The long-term coastal operations were categorized into two distinct phases. Stage I: startup cycling. During the initial 75-hour period (3 cycles), 23.76 L of seawater underwent batch-cyclic electrolysis with concurrent air sparging. This phase established the requisite alkalinity and achieved high ion removal efficiencies, 95.94% for Ca^{2+} and 79.52% for Mg^{2+} (Figure 6b-c), while stabilizing the pH at 10.16 (Figure 6d), with an H_2 production rate of 935 mL h^{-1} (Figure S18a-b). Stage II, steady-state continuous flow. Following startup, the system transitioned to a steady-state flow of 330 mL h^{-1} with concurrent air injection for CO_2 capture. Effluent Ca^{2+} concentrations stabilized at $12.56 \pm 1.15 \text{ mg L}^{-1}$ (96.8% removal), while Mg^{2+} levels stabilized at $318.62 \pm 62.37 \text{ mg L}^{-1}$ (73.6% removal). The slight decrease in Mg^{2+} removal efficiency relative to the batch phase (79.52% in stage I) is attributed to local OH^- consumption due to continuous saturated CO_2 injection,

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which slightly shifts the pH away from the optimal $\text{Mg}(\text{OH})_2$ precipitation window. This observation highlights a nuanced interplay between CO_2 mass transfer and OH^- availability, providing a clear target for future optimization of dynamic control.

The system's robustness was rigorously verified through a 250-hour continuous-flow campaign (10 cycles). During this period, the PV-SE system processed 79.2 L of natural seawater (Note S3), recovering 229.73 g of mixed $\text{Mg}(\text{OH})_2$ and CaCO_3 precipitates (Figures 6g and S18c), corresponding to a solid-phase mineral recovery rate of $2.21 \text{ kg}^{-1} \text{ m}^{-2} \text{ day}^{-1}$ per unit electrode area. Effective CO_2 sequestration in the liquid phase was confirmed by a significant increase in effluent TC from 40.1 to 181.5 mg L^{-1} (Figure S18d). Simultaneously, the anode produced disinfectant-grade free chlorine at 870.3 mg L^{-1} (Figure 6h) during a single anodic electrolysis cycle, exceeding the lower threshold for disinfectant standards. Mass and charge balance calculations over long-term cycling indicated recovery efficiencies of 75.56% for calcium and 73.71% for magnesium, alongside a CO_2 capture efficiency of 73.14% (Note S4 and Table S18). The liquid-phase free chlorine stabilization efficiency was quantified at 65.88%, which can be further enhanced by alkaline conversion to stable NaClO , thereby mitigating potential environmental risks. Over the 250-hour duration, the system exhibited exceptional stability, with an average electrolysis voltage of 2.66 V, which was remarkably consistent with the lab-scale single-channel results (2.64 V). The recovery voltage (-2.86 V , Figure 6e) remained stable, while Fourier transform infrared (FTIR) spectroscopy and SEM images confirmed the structural integrity of the AEM post-operation (Figure S19). Critically, the PV-battery integration ensured complete energy autonomy, maintaining a self-sustaining closed-loop supply throughout the diurnal fluctuations in solar output (Figure 6f). Collectively, these results substantiate the commercial feasibility of using PV-driven seawater electrolysis as a scalable framework for sustainable resource recovery and global carbon management.

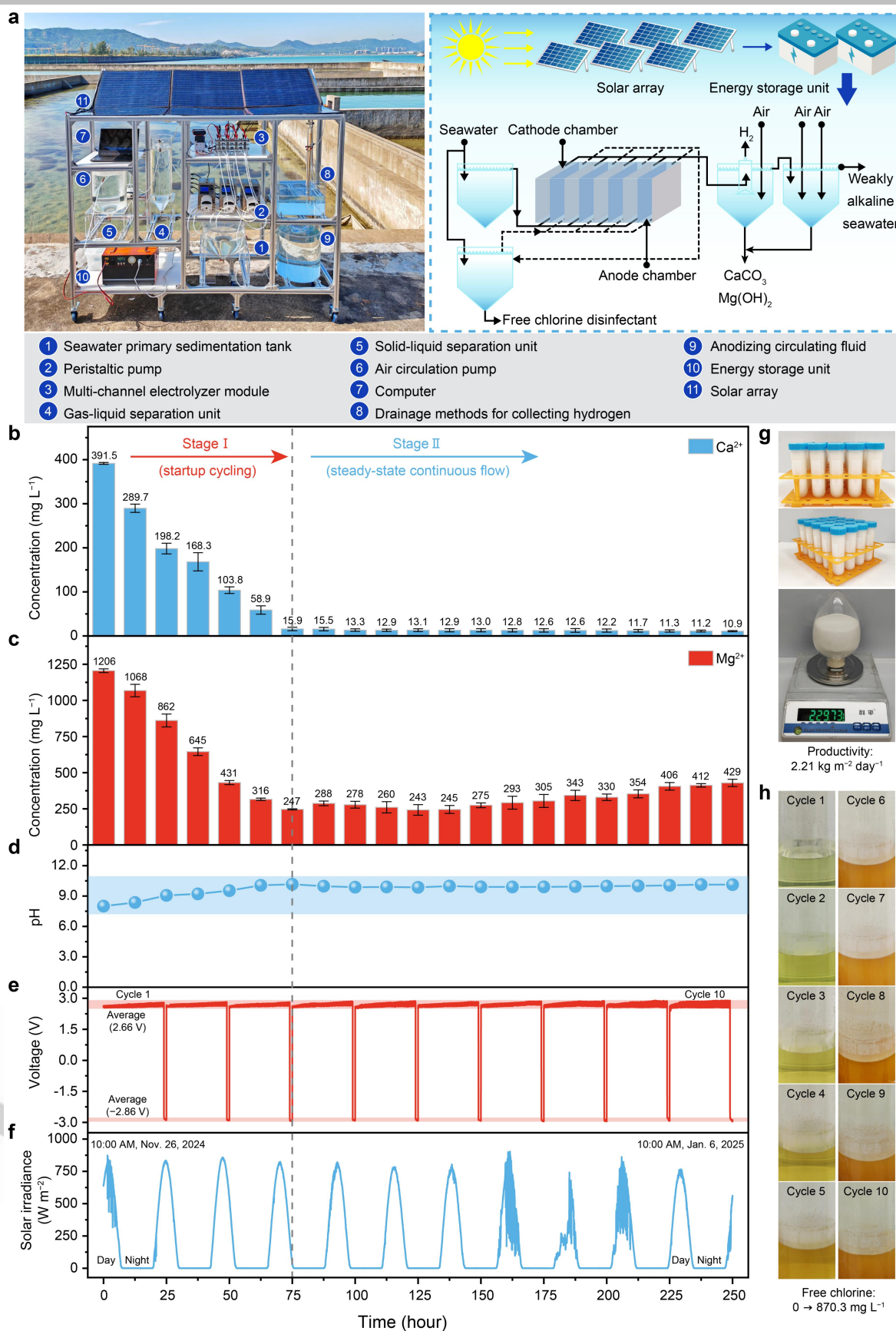


Figure 6. Operational performance of the integrated PV-SE system under coastal field conditions. (a) Optical image and comprehensive process flow diagram of the integrated PV-SE system deployed at Shenzhen Dapeng Bay. (b) Effluent concentrations of Ca²⁺ and (c) Mg²⁺ from the cathode chamber during continuous operation at 10 mA cm⁻² and a RHRT of 20 min. Temporal profiles of (d) system pH, (e) voltage, and (f)

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solar irradiance fluctuations during 250 h of continuous-flow operation. (g) Optical image of calcium and magnesium precipitates recovered from the cathode chamber. (h) Visualization of the disinfectant-grade chlorine solution generated in the anode chamber (exceeding 100 mg L⁻¹ available chlorine, Chinese GB28233-2020 standard). Error bars represent means $\pm$ standard deviation of three replicates.

Total Carbon Capture Capacity and Techno-Economic Analysis

To evaluate the cumulative carbon mitigation potential of the PV-SE system, we rigorously quantified the total carbon capture capacity using a comprehensive four-pathway mineralization pathway^[34]. This analysis accounts for the synergistic contributions of both dual solid-phase (CaCO₃ and Mg(OH)₂) and dual liquid-phase (TC and pH) routes (Note S5). Specifically, the Ca²⁺ fraction in seawater facilitated the direct precipitation of CaCO₃, achieving a carbon capture capacity of 324.49 g CO₂ m⁻³ (Tables S7-S10). Concurrently, the recovered Mg(OH)₂ provides a significant reservoir for indirect carbonation, with the potential to sequester 1617.89 g CO₂ m⁻³ (Figure S20 and Table S11), through the formation of MgCO₃. Liquid-phase capture, driven by increases in electrolyte TC and pH, contributed an additional 518.18 and 4.18 g CO₂ m⁻³, respectively (Tables S12, S13). By simultaneously maximizing mineral resource valorization and atmospheric CO₂ absorption, the PV-SE system achieved a record total capture capacity of 2,464.74 g CO₂ m⁻³ of processed seawater (Figure 7a and Table S14). Notably, solid-phase precipitation accounted for 78.8% of this capacity, while liquid-phase alkalinity enhancement contributed 21.2% (Figure 7b). This establishes that the total carbon capture capacity exceeds that of state-of-the-art electrochemical capture technologies (Table S15), making the PV-SE system an exceptionally efficient solution for sustainable carbon management.

The economic and operational viability of the PV-SE system was further substantiated through a detailed Techno-Economic Analysis (TEA). The system exhibited remarkable energy autonomy, utilizing only 61% of its PV-battery output (Figure 7c). Electrolysis for CO₂ capture required a minimal energy investment of 3.44 kWh kg⁻¹, representing only 4.0% of the total system energy budget (Table S16), a figure that could be further reduced by employing highly efficient catalytic electrode materials^[46,47]. Pumping (67.5%) and data acquisition (22.5%) were identified as the primary energy consumers and were targeted for further engineering optimization^[38]. This efficient allocation yields a 39% energy surplus (Note S6), enabling uninterrupted autonomous operation for 250 hours. To optimize system economics, the replacement of Pt/C and RuO₂ with low-cost alternatives was experimentally verified (Figure S21). Although some non-precious catalysts exhibited slightly higher overpotentials than commercial benchmarks, their minimal material cost drastically reduces future scale-up capital expenditure. Crucially, optimizing porous nickel felt electrodes yielded an overpotential reduction exceeding 23.42%. This performance gain stems from the Ni felt's microporous architecture, which accelerates gas bubble release and expands the electrochemically active surface area (ECSA), offering strategic insights for noble-metal-free electrode construction within seawater mineralization.

From an environmental perspective, the alkaline supernatant (pH = 9.29) effluent continues to absorb atmospheric CO₂ upon discharge into an open-air environment, naturally equilibrating to a safe, weakly alkaline seawater level (pH = 8.54) within 10 days (Figure S22a) and thereby effectively mitigating localized ecological risks. Given that active residual oxidants and dissolved catalyst-associated heavy metal ions remain undetected (Figure S22b and Table S17), the immediate environmental risks associated with discharging the catholyte under the PV-SE system are minimized. Nevertheless, more comprehensive long-term assessments of biological ecotoxicity and chronic environmental impacts will be indispensable before any future scale-up of this process. Economically, the system achieves a gross profit of \$104.12 per ton of CO₂ captured (excluding potential CO₂ credits) through the sale of co-produced H₂, disinfectants, and Ca/Mg nanomaterials (Figure 7d and Note S7). This estimate is conservative and well below standard market benchmarks, and co-product sales are intended primarily to offset operational costs and sustain the closed-loop system, whereas future revenue from carbon credits is expected to be the primary long-term profit driver. This potential profitability, coupled with the potential for efficiency gains via advanced catalytic electrodes and a scaled reactor in the future, underscores the PV-SE system's readiness for industrial-scale deployment as a cornerstone of coastal carbon neutrality (Table S19).

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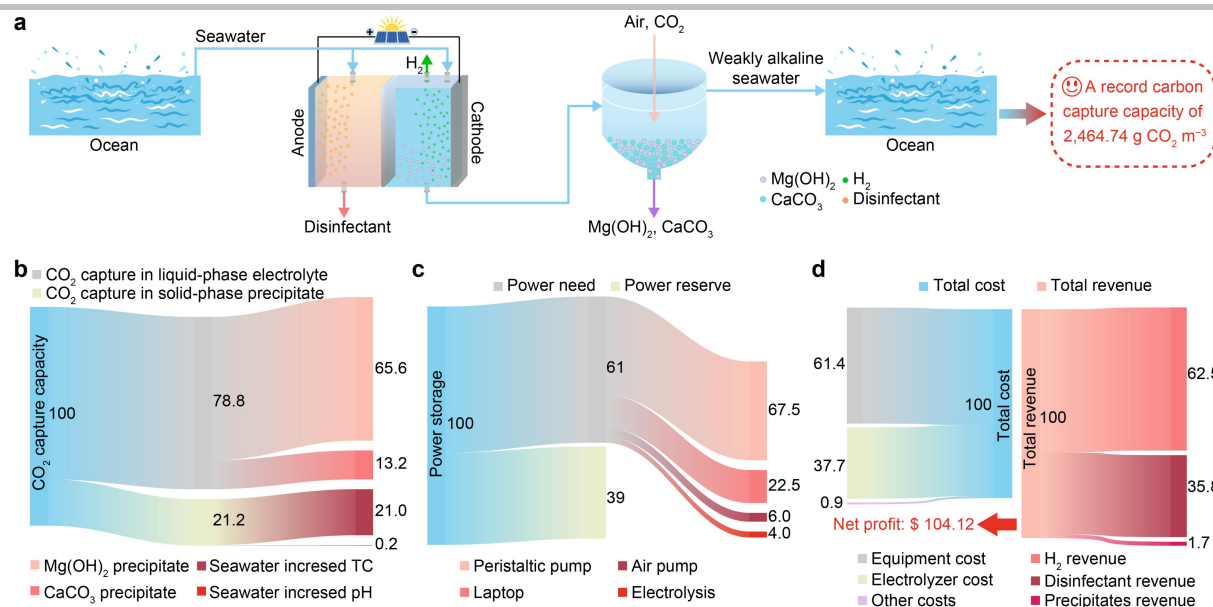


Figure 7. Quantification of carbon capture capacity and techno-economic feasibility. (a) Integrated material flow and mass balance of the PV-SE system, illustrating the nexus between seawater resource recovery and CO_2 sequestration. (b) Relative contributions of dual solid-phase (precipitates) and dual liquid-phase (electrolyte) pathways to the total carbon capture capacity. (c) Energy distribution profile identifying the power consumption of primary system components and the total energy surplus. (d) Comprehensive techno-economic breakdown detailing operational expenditures and revenue streams from co-products.

Conclusion

In summary, this study introduces and validates the photovoltaic-driven seawater electrolysis (PV-SE) system, a robust and integrated framework designed to address the critical nexus between resource scarcity and atmospheric CO_2 capture. The PV-SE system harnesses renewable solar energy to power the cathodic HER and generate OH^- , thereby enabling the concurrent co-production of green H_2 and disinfectant-grade free chlorine. Through a combination of in situ and ex situ characterization, we have elucidated the dynamic interplay between cathodically generated alkalinity and the precipitation kinetics of Ca^{2+} and Mg^{2+} . Critically, we demonstrate that the competition between $Mg(OH)_2$ formation and CO_3^{2-} generation is precisely tunable by the partial pressure of CO_2 . This insight establishes a novel, elegant, and highly controllable strategy for the selective recovery of phase-pure, food-grade $Mg(OH)_2$ (99.43%) and $CaCO_3$ (99.78%) resources from complex marine matrices. The engineering viability of the integrated PV-SE system was substantiated by a 250-hour, energy-autonomous pilot study that achieved a record-breaking carbon capture capacity of $2,464.74 \text{ g CO}_2 \text{ m}^{-3}$, alongside substantial mineral recovery. Our techno-economic analysis further underscores the commercial potential of this technology, projecting that byproduct revenue from per tonne of CO_2 captured can fully offset baseline operating expenses, while future revenue from carbon credits is positioned to serve as the primary driver of long-term profitability. The PV-SE system represents a significant step forward in coupled resource-climate solutions. By successfully integrating sustainable energy harvesting, high-efficiency carbon sequestration, and high-value resource valorization, the PV-SE system, as a fully closed-loop energy supply, provides a scalable and economically viable framework for coastal carbon management. This work paves the way toward achieving global carbon neutrality goals through the innovative and sustainable utilization of marine resources.

Supporting Information

The authors have cited additional references within the Supporting Information [48–52].

Acknowledgements

This work was supported by the National Key Research and Development Program of China (Grant No. 2021YFA1202500); the Shenzhen Key Laboratory of Interfacial Science and Engineering of Materials (No. ZDSYS20200421111401738); the Stable Support Plan Program of Shenzhen Natural Science Foundation (No. 20231122110855002); Shenzhen Science and Technology Program (No. JCYJ20241202123900001, JCYJ20241202125900002); Guangdong Basic and Applied Basic Research Foundation (No. 2023A1515110125);

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High-level University Special Fund (G03050K001). We acknowledge the technical support from the SUSTech Core Research Facilities at the Southern University of Science and Technology.

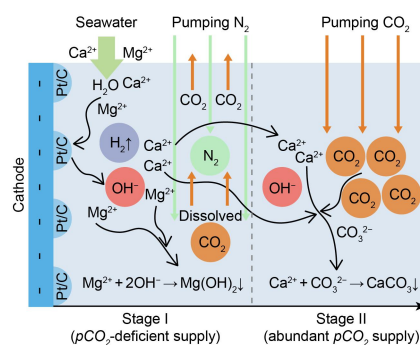
Keywords: electrochemical • seawater electrolysis • carbon capture • resource recovery • photovoltaic-driven

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By integrating sustainable energy harvesting, high-efficiency carbon sequestration, and high-value marine resource valorization through a CO₂ partial pressure (*p*CO₂)-steering mineralization mechanism, the photovoltaic-driven seawater electrolysis (PV-SE) system offers a scalable, economically viable framework for coastal carbon management and a pivotal pathway for sustainable valorization of marine resources.